

Representative Structural Calculation Package

Apartment frame equivalent

Item	Value
Project	Apartment frame equivalent
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	07-apartment-frame-equivalent

Nodes	Members	Loads	Combos	Max disp.	Max util.
126	270	756	28	4.173 mm	0.358

Model And Design Basis

Item	Value
Building type	apartment
Description	Six-story elongated residential block represented as an equivalent 3D frame.
Occupancy	Residential
Bounds X/Y/Z	25.00 / 7.600 / 17.40
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	162	0.000 kN	0.000 kN	-3420 kN
L	live	162	0.000 kN	0.000 kN	-1482 kN
WX	wind	108	59.51 kN	0.000 kN	0.000 kN
WY	wind	108	0.000 kN	195.75 kN	0.000 kN
EX	seismic	108	344.94 kN	0.000 kN	0.000 kN
EY	seismic	108	0.000 kN	344.94 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	190.00	570.00 kN	266.00 kN	9.918 kN	32.62 kN
2	190.00	570.00 kN	266.00 kN	9.918 kN	32.62 kN
3	190.00	570.00 kN	266.00 kN	9.918 kN	32.62 kN
4	190.00	570.00 kN	266.00 kN	9.918 kN	32.63 kN
5	190.00	570.00 kN	266.00 kN	9.918 kN	32.63 kN

6	190.00	570.00 kN	152.00 kN	9.918 kN	32.62 kN
---	--------	-----------	-----------	----------	----------

Load Derivation Formula Trace

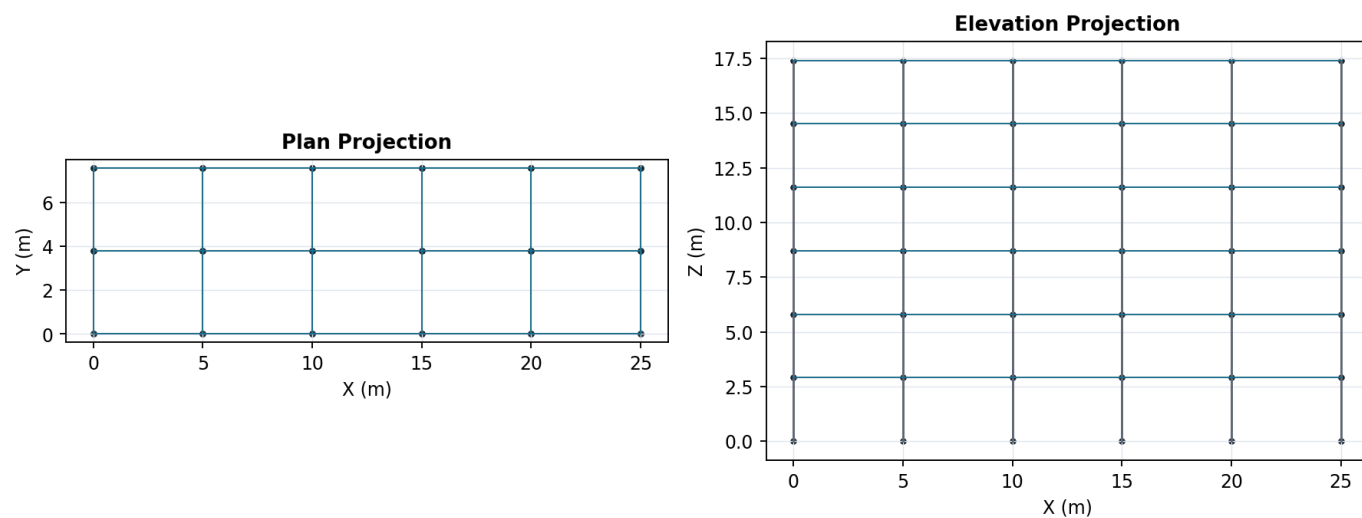
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=residential
basis-plan-area	-	max(model width * model depth, 1)	190.00 m2	Bx=25.00 m, By=7.600 m
basis-roof-area	-	roofArea input or typical floor area	190.00 m2	roofArea=190.00 m2
D-ST1	D	$A * q_D$	570.00 kN	$A=190.00 \text{ m}^2$, $q_D=3.000 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	266.00 kN	$A=190.00 \text{ m}^2$, $q_L=1.400 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	9.918 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=7.600 \text{ m}$, $h=2.900 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	32.62 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=25.00 \text{ m}$, $h=2.900 \text{ m}$
D-ST2	D	$A * q_D$	570.00 kN	$A=190.00 \text{ m}^2$, $q_D=3.000 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	266.00 kN	$A=190.00 \text{ m}^2$, $q_L=1.400 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	9.918 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=7.600 \text{ m}$, $h=2.900 \text{ m}$
WY-ST2	WY	$p_{WY} * B_x * h$	32.62 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=25.00 \text{ m}$, $h=2.900 \text{ m}$
D-ST3	D	$A * q_D$	570.00 kN	$A=190.00 \text{ m}^2$, $q_D=3.000 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	266.00 kN	$A=190.00 \text{ m}^2$, $q_L=1.400 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	9.918 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=7.600 \text{ m}$, $h=2.900 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

Apartment frame equivalent



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	0.763 mm	0.088	0.000
KDS-ST-02	OK	0.995 mm	0.119	0.000
KDS-ST-04-WX-P	OK	1.017 mm	0.116	0.000
KDS-ST-04-WX-N	OK	1.017 mm	0.116	0.000
KDS-ST-04-WY-P	OK	2.025 mm	0.149	0.000
KDS-ST-04-WY-N	OK	2.025 mm	0.149	0.000
KDS-ST-05-EX-P	OK	3.928 mm	0.182	0.000
KDS-ST-05-EX-N	OK	3.928 mm	0.182	0.000
KDS-ST-05-EY-P	OK	4.173 mm	0.187	0.000
KDS-ST-05-EY-N	OK	4.173 mm	0.187	0.000
KDS-ST-06-WX-P	OK	0.721 mm	0.070	0.000
KDS-ST-06-WX-N	OK	0.721 mm	0.070	0.000
KDS-ST-06-WY-P	OK	1.895 mm	0.103	0.000
KDS-ST-06-WY-N	OK	1.895 mm	0.103	0.000
KDS-ST-07-EX-P	OK	3.858 mm	0.136	0.000
KDS-ST-07-EX-N	OK	3.858 mm	0.136	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	0.904 mm
Max drift ratio	0.00031
Governing	KDS-ST-05-EY-N / story 2

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	2.900 m	0.005 mm	0.00000	0.000	OK
KDS-ST-01	2	2.900 m	0.004 mm	0.00000	0.000	OK
KDS-ST-01	3	2.900 m	0.000 mm	0.00000	0.000	OK
KDS-ST-01	4	2.900 m	0.000 mm	0.00000	0.000	OK
KDS-ST-01	5	2.900 m	0.003 mm	0.00000	0.000	OK
KDS-ST-01	6	2.900 m	0.023 mm	0.00001	0.002	OK
KDS-ST-02	1	2.900 m	0.007 mm	0.00000	0.000	OK
KDS-ST-02	2	2.900 m	0.005 mm	0.00000	0.000	OK
KDS-ST-02	3	2.900 m	0.000 mm	0.00000	0.000	OK
KDS-ST-02	4	2.900 m	0.000 mm	0.00000	0.000	OK
KDS-ST-02	5	2.900 m	0.003 mm	0.00000	0.000	OK
KDS-ST-02	6	2.900 m	0.027 mm	0.00001	0.002	OK
KDS-ST-04-WX-P	1	2.900 m	0.104 mm	0.00004	0.007	OK
KDS-ST-04-WX-P	2	2.900 m	0.138 mm	0.00005	0.009	OK
KDS-ST-04-WX-P	3	2.900 m	0.114 mm	0.00004	0.008	OK
KDS-ST-04-WX-P	4	2.900 m	0.087 mm	0.00003	0.006	OK
KDS-ST-04-WX-P	5	2.900 m	0.061 mm	0.00002	0.004	OK
KDS-ST-04-WX-P	6	2.900 m	0.057 mm	0.00002	0.004	OK
KDS-ST-04-WX-N	1	2.900 m	0.104 mm	0.00004	0.007	OK
KDS-ST-04-WX-N	2	2.900 m	0.138 mm	0.00005	0.009	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M100	OK	0.358	steel-deflection	KDS-ST-05-EY-N
2	M99	OK	0.358	steel-deflection	KDS-ST-05-EY-P
3	M94	OK	0.358	steel-deflection	KDS-ST-05-EX-N
4	M106	OK	0.358	steel-deflection	KDS-ST-05-EX-N
5	M93	OK	0.358	steel-deflection	KDS-ST-05-EX-P
6	M105	OK	0.358	steel-deflection	KDS-ST-05-EX-P
7	M101	OK	0.358	steel-deflection	KDS-ST-05-EY-P
8	M98	OK	0.358	steel-deflection	KDS-ST-05-EY-N
9	M95	OK	0.358	steel-deflection	KDS-ST-05-EX-P
10	M107	OK	0.358	steel-deflection	KDS-ST-05-EX-P

11	M104	OK	0.358	steel-deflection	KDS-ST-05-EX-N
12	M92	OK	0.358	steel-deflection	KDS-ST-05-EX-N
13	M102	OK	0.355	steel-deflection	KDS-ST-05-EY-N
14	M97	OK	0.355	steel-deflection	KDS-ST-05-EY-P
15	M108	OK	0.355	steel-deflection	KDS-ST-05-EY-P
16	M96	OK	0.355	steel-deflection	KDS-ST-05-EY-N
17	M103	OK	0.355	steel-deflection	KDS-ST-05-EY-P
18	M91	OK	0.355	steel-deflection	KDS-ST-05-EY-N
19	M76	OK	0.327	steel-deflection	KDS-ST-05-EX-P
20	M88	OK	0.327	steel-deflection	KDS-ST-05-EX-P

Member Design Trace Matrix

Item	Value
Checked members	270
Unimplemented members	0
NG / WARN	0/0
Max utilization	0.358

Member	Type	Status	Util.	Formula	Action
M100	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M99	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M94	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M106	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M93	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M105	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M101	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M98	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M95	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M107	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

M104	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M92	steel	OK	0.358	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M102	steel	OK	0.355	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M97	steel	OK	0.355	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M108	steel	OK	0.355	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M96	steel	OK	0.355	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M103	steel	OK	0.355	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M91	steel	OK	0.355	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M76	steel	OK	0.327	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M88	steel	OK	0.327	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	270	0	0.358
Connections	270	54	1.842
Foundations	18	54	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.